Lesson plan · for the teacher

# What is data science?

Data science · Level 1 · Lesson 1

**Time**
40 min

**Materials**
Board and chalk or markers; Last month's register; A phone or tablet with Darsi, for the lab

## Learning goals

- Say what data science is.
- Follow the loop: ask, collect, explore, explain.
- Turn a vague question into a measurable one.

## Activity: From a guess to an answer

1. Warm-up (5 min): Ask for guesses about why pupils are late and write them on the board. Point out that these are guesses, not answers.
2. Teach (8 min): Draw the loop: ask, collect, explore, explain. Use the register as the data, and show how a pattern by day leads to a new question.
3. Count (15 min): Groups take one week each of last month's register (no names, just counts) and tally late arrivals by day. Combine the groups' counts on the board in one table.
4. Explain (7 min): Draw a bar chart from the table. Ask: which day stands out? What could explain it? What should we count next?
5. Wrap-up (5 min): Ask two pupils to say the finding in one sentence, with a number in it. Then give the exit ticket.

**Use counts only; never show which pupil was late.**

## Discussion

- Which guess on the board did the data support?
- What would we need to count to check the bus idea?
- Which school question would you ask next?

## Exit ticket

- Write one measurable question about our school, and the record that would answer it.

**Try it in class tomorrow**
Ask your pupils for questions about the school that data could answer. Choose one you can measure, and plan what to count this week.

Worksheet · for pupils

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Name _____ Class _____ Date _____

**1** **We explained our finding. What comes next in the loop?**

◯ A new question ◯ Stop: we are done ◯ Delete the data

**2** **Data scientists mainly build the databases that store a school's data.**

◯ True ◯ False

**3** **Put the data science loop in order.**

☐ Explore with numbers and a chart

☐ Ask a clear question

☐ Explain the finding, then ask again

☐ Collect the data it needs

**4** **The list holds the days on which pupils arrived late last week. What does this program print?**

```
late = ["Mon", "Tue", "Mon", "Wed", "Mon", "Fri", "Mon"]
monday = 0
for day in late:
    if day == "Mon":
        monday = monday + 1
print(monday)
```

◯ Mon ◯ 4 ◯ 1 ◯ 7

**5** **Match each school question to the data that answers it.**

| | | |
|---|---|---|
| Which days do pupils arrive late? | • • • • | Library cards |
| Which books are borrowed most? | • • • • | The register |
| Which month is wettest? | • • • • | Marks for each question |
| Which topic was hardest in the test? | • • • • | The rain gauge record |

Answers · for the teacher

# What is data science?

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**1** **A new question**

Data science is a loop: every answer leads to the next question, as the Monday pattern led to the bus.

**2** **False**

That is the data engineers' job. Data scientists ask the data questions, find patterns and explain them.

**3** **Top to bottom: 3, 1, 4, 2**

The question comes first, so you collect only what you need.

**4** **4**

The loop looks at each day in turn, and the if adds 1 only for a Monday: 4 of the 7 late arrivals.

**5** **Which days do pupils arrive late? — The register**
**Which books are borrowed most? — Library cards**
**Which month is wettest? — The rain gauge record**
**Which topic was hardest in the test? — Marks for each question**

Each good question points to data the school already keeps.

✎ **For you to think about**

**What is one question about your school that you would like data to answer? Where is the data?**